Waiz Khan

(443) 378-2030 | wkhan12@jh.edu | LinkedIn | GitHub | Portfolio

Education

Johns Hopkins University — Baltimore, MD
M.S.E. in Data Science — *Aug. 2024 – May 2027*
- Coursework: Machine Translation, Deep Learning, Artificial Intelligence, Applied Statistics and Data Analysis, Introduction to Pattern Recognition, Introduction to Data Science

GD Goenka University — Gurugram, India
B.Tech. in Computer Science & Engineering — *Aug. 2019 – Nov. 2023*

Publications

W. Khan and P. Koehn. Measuring the Language Share of Multilingual Representations. *Under review at ICLR 2027.*

Research Experience

Research Assistant — Dec. 2025 – Present
Johns Hopkins University, Advisor: Prof. Philipp Koehn — *Baltimore, MD*

How Multilingual Models Separate Language from Meaning — *First author; Aim 1 of an NSF proposal*
- Introduced Language Factor Share (LFS), a layerwise two way variance decomposition that measures how much of a model's internal representation reflects language identity rather than sentence meaning; applied it to 17 open models (0.6B to 9B parameters) across 128 languages on a Slurm cluster
- Found that every model shows the same shape: language information falls in the middle layers and returns near the output (17 of 17, replicated on FLORES-200 with rank correlation 0.92); the depth of the dip varies nine fold and is not explained by model size alone
- Showed the shared middle is not English: an average of the other languages predicts each language better than English does for 113 to 124 of 127 languages, and 63 to 97% of cross language misalignment is a simple offset plus a scale
- Validated before claiming: 31 preregistered predictions (21 passed) and 4 of 4 blind predictions on unseen model families; a smaller language share tracks more use of foreign language context (Spearman 0.51)
- Tested LFS as a training objective and showed it fails: optimizing it doubled the dip in 400 steps while Belebele accuracy fell (2.9 points vs. 1.7 for the control), so the paper reports it as a measurement, not a loss

Ancient Script Decipherment
- Built a leakage controlled benchmark of 12 ancient script corpora and sign inventories with 29 frozen evaluation partitions; an audit found roughly 34% of the Egyptian evaluation data duplicated from training
- Recovered how Egyptian signs function from context and position alone, with no translations, dictionary, or labels, above chance on the cleaned data (ARI 0.169, $p = 0.0015$)

Architectures for Learning Language from Little Data
- Ran 43 experiments comparing decoder architectures (Transformer, Mamba, mixture of experts, test time training) for Egyptian translation; rankings flipped when input moved from sign IDs to images, with test time training the most robust; traced promising local learning rule results to a pooling bug and withdrew them

Independent Research — Aug. 2026 – Present
Episodic Memory for Continual World Models — *Baltimore, MD*
- Built a prototype comparing event based and moment based episodic memory; found part of the event memory's advantage came from reaching further back in time, not from its structure
- Designed matched tasks (same current view, different histories, different correct actions) to test whether memory changes a world model's plan; proposal written for Prof. Randall Balestriero's group at Brown

Research Assistant — Aug. 2022 – Nov. 2022
GD Goenka University, Advisor: Dr. Yogesh Kumar | extended independently in 2026 — *Gurugram, India*
- Predicted chronic school absenteeism on 47,027 students from IHDS-II national survey data, comparing 8 models on splits grouped by state (ROC-AUC 0.841 vs. 0.699 for logistic regression) and reporting ranking and calibration separately; tested distance to school as a cause with DoWhy refutation checks

Experience

Co-Founder & CTO — Apr. 2026 – Present
Iris, healthcare AI | JHU Pava Center Accelerator, Spark Program | ARPA-H ADVOCATE 2026 proposal — *Baltimore, MD*
- Built a versioned patient record with provenance on every value, and automated checks that block any AI generated claim that cannot be traced back to source records

Co-Founder & Vice Chair, *Heela (heela.org), nonprofit for college access for refugee students* — 2026 – Present

Technical Skills

Research Methods: representation analysis, variance decomposition, preregistration, blind held out evaluation, leakage audits, falsification tests, permutation testing, causal inference
Tools: Python, PyTorch, Hugging Face, scikit-learn, SQL, TypeScript, C++, Bash, Linux, Slurm HPC, Git, Docker